

Starter Project #3

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For this assignment, we were asked to create a class within a pre-made program called Boggle, a word search puzzle game. Letters are arranged on a grid, and the purpose of the game is to find as many words as possible by connecting adjacent letters horizontally, vertically, or diagonally. After I had time to learn the game myself, I began coming up with an approach to creating, developing, and testing the game.

After understanding the game, I began brainstorming the most efficient way to get this done. Soon, I figured out how I wanted to approach the start of the program. I wrote and set up the program by initializing all data structures that I'd need. Then, I broke the program up into 4 different methods, and multiple sections focusing on grid validation, dictionary validations, fast dictionary, and many extra edge cases. Throughout this process, I utilized my resources such as office hours, peers, online websites, and LLMs to provide assistance on parts of the program that I wasn't confident in. For example, in class, Professor Burge broke down what a hash map is. After still being a little confused, I used the internet and what I know to break down the hash map and program it. After going to the office hours, I was informed by professor burdge and the TA Howard that my hash map could be improved, and I got assistance from them.

During peer review, my reviewer gave me good feedback, mostly based on my formatting and readability. During my time programming the game, I strategically placed print statements throughout my code to help identify errors and bugs. This is a method I often use to figure out which part of my program is failing. When she reviewed the code, she noticed these print statements and suggested I remove them now that we know the program works. This cleans up the output of the program and removes the extra lines of code. Another comment she left was based on the comments in my code.

My peer reviewers' program was very easy to read and follow along with. It was well organized, utilizing comments, indentation, and more. Her hash map was very effective and

executes stopping unnecessary searches early. When reviewing her program, at the top of the class, I noticed two methods that were created, but as I scrolled through, they were never used or called in the class. So, my main feedback was for her to remove those two unnecessary methods to improve the overall efficiency and cleanliness.

During the creation of the program, I made major improvements. As I mentioned above, my Hash Map was my biggest improvement I made because I actually changed the structure of the Hash Map. Next, would have to be the way I implemented my checks and edge cases. At first, my checks were really random and unordered, and had a lot of extra unnecessary lines of code in my program. This caused my program to run less smoothly and made my code not look appealing. So, I corrected this by organizing my code and using loops to check for multiple things at once. I also used some Boolean and loops to decrease the lines of code in my program.

After all my implementations, it was time to test my code and make sure it runs and passes every test case. Now, this part was a little challenging because my code wouldn't produce any output for a long time. I used the method I mentioned earlier and added print statements after specific parts of my code. I did this to test where exactly my program is stopping or failing. After doing this for a while, I figured out what the problem was. It was a quick and simple fix, allowing me to quickly make my adjustments. Lastly, I tested my code with multiple different dictionaries and grids, then when I felt confident in my ability to pass the test, I checked it and passed all tests.

The project taught me a lot about how professional workspaces run. From the implementation of our Boggle class, creating our unit tests, and code reviews, this project gave me another experience that I haven't had the opportunity to understand and experience. I was able to understand and create my own unittest to test the correctness of my code and test random edge cases. Then I learned how to push my code to github allowing my peer reviewer to give me feedback that I was able to implement to improve my code.

